Islamic University of Technology

Lab Report

Experiment No: 04

Student No: 210041105

Course Code: CHEM-4242

Name of the Experiment: ESTIMATION OF COPPER CONTAINED IN A SUPPLIED SOLUTION BY IODOMETRIC METHOD

Date of Performance: 10/3/23

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Submitted by:

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Section: 01 (A)

Dept. of Computer Science and Engineering (CSE)

Objective:

To standardize the sodium thiosulphate solution by titration method and to determine the mass of unknown copper sample using iodometric method.

Theory:

Cu^{2+} is iodometrically estimated by treating with an excess of KI solution and titrating the liberated iodine with a sodium thiosulphate ($Na_2S_2O_3 . 5H_2O$) solution which is standardized against a standard $K_2Cr_2O_7$ solution.

$$2Cu^{2+} + 4I^- \longrightarrow 2CuI\downarrow + I_2$$

$$I_2 + S_2O_3^{2-} \longrightarrow 2I^- + S_4O_6^{2-}$$

From the chemical reactions, we can derive:

$\therefore$ 2 moles of Cu^{2+} ≡ 1 mol I_2 ≡ 2 moles of $S_2O_3^{2-}$

$\therefore$ 1 mol of $S_2O_3^{2-}$ ≡ 1 mol Cu^{2+} ≡ 1 equivalent

$\therefore$ 1000 mL 1N $Na_2S_2O_3$ solution ≡ 63.55 gm Cu^{2+}

Sodium thiosulphate solution is standardized against a standard $K_2Cr_2O_7$ solution.

$$Cr_2O_7^{2+} + 14H^+ + 6I^- \longrightarrow 2Cr^{3+} + 3I_2 + 7H_2O$$

$$I_2 + S_2O_3^{2-} \longrightarrow 2I^- + S_4O_6^{2-}$$

Chemicals required:

1. Standard N/20 $K_2Cr_2O_7$ solution.
2. $Na_2S_2O_3.5H_2O$ solution.
3. 15% KI solution
4. 4 N H_2SO_4.
5. Freshly prepared 1% starch solution

Apparatus required:

1. Conical flask
2. Pipette
3. Burette.
4. Funnel
5. Stand and clamp.
6. Watch glass.

Procedure:

We pipette out 10 mL of supplied copper salt solution into a conical flask. We add a few drops of diluted Na_2CO_3. A pale greenish precipitate is observed. The precipitate is then dissolved by adding few drops of acetic acid (CH_3COOH). Then, we add about 10 mL of 10% potassium iodide (KI) solution and titrate the liberated iodine against the standard thiosulphate solution. This titration is carried out until the brown colour of iodine changes to light yellow. Next, 1-2 mL of starch solution is added and titration is continued until the blue colour begins to fade. After that, few drops of 10% ammonium thiocyanate solution are added and titration is continued until the blue colour is just discharged. We finally calculate the amount of copper present in one litre of the supplied solution.

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A (CSE)

EXP-4:

ESTIMATION OF COPPER CONTAINED IN A SUPPLIED SOLUTION BY IODOMETRIC METHOD

10/3/23

Experimental Data:

No. of Obs.	Volume of Cu^{2+} solution taken (mL)	Burette Reading for $Na_2S_2O_3$ soln (mL)		Diff. of burette Reading	Avg. volume of $Na_2S_2O_3$ soln required (mL)
		Initial	Final		
1	10.0	0.2	9.74	9.54	9.57
2		9.74	19.31	9.57	
3		19.31	28.92	9.61	

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Data Calculation:

Supplied,

Concentration of $Na_2S_2O_3$ solution = 0.103 N

Concentration of Cu^{2+} in supplied solution = 6.28 g/L

1 mL 1N $Na_2S_2O_3 \equiv 0.06354$ g of Cu^{2+}

$\therefore$ Amount of $Cu^{2+} = 0.06354 \times 0.103 \times 9.573$ gm

$= 0.0626$ gm

$\therefore$ Concentration of $Cu^{2+} = \dfrac{0.0626 \times 1000}{63.5 \times 10}$

$= 0.0986$ N

Here, Concentration of $Cu^{2+} = 0.0986 \times 63.5$

$= 6.26$ gm/L

10/7/22

Error Calculation:

$$\text{Error} = \frac{6.28 - 6.26}{6.28} \times 100\%$$

$$= 0.318\%$$

Result:

Concentration of Cu^{2+} is 6.26 gm/L

Discussion:

In this experiment, we note the presence of multiple reagents and hence, we must be careful while performing each individual steps or chemical reactions. This experiment familiarizes us with the process of iodometry since we see during the titration, the yellowish colour of iodine changes drastically to a bluish grey precipitate. The analysis of the experiment shows us that there was a 0·318% error. This error could have occured due to human error while operating the burette or during calculation.